# Omar Younis

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## Education

**California State University, Fullerton** — Fullerton, CA
*Master of Science in Computer Science* — *Aug. 2023 – May 2026*

- M.S. Project: GPU parallelization of Brent's root-finding method in CUDA — 35x kernel-level and 8.8x end-to-end speedup on an NVIDIA RTX 3080.

**Worcester Polytechnic Institute** — Worcester, MA
*Bachelor of Science in Mechanical Engineering* — *Aug. 2012 – May 2016*

- Concentration in Design; Minor in Aerospace Engineering

## Experience

**Software Engineer Intern** — May 2024 – Aug. 2024
*Qualcomm* — *San Diego, CA*

- Built and industrialized ETL pipelines on AWS, translating requirements from cross-functional stakeholders into production data flows processing 1M+ data points per minute for ML teams.
- Cut feature development and testing time by 40% by adding logging, monitoring, and alerting across the pipeline stack.

**Instructor / Teaching Associate – Computer Science** — Jan. 2024 – May 2024
*California State University, Fullerton* — *Fullerton, CA*

- Taught an undergraduate Object-Oriented Programming course in C++; built course materials and coached students on software-engineering practices.

**Machine Learning Engineer** — Aug. 2022 – Nov. 2022
*Elemeno AI* — *Remote*

- Trained a feed-forward neural network to predict package delivery times, improving predictive accuracy by 25% through data analysis and tabular-data pipeline design.
- Ran exploratory data analysis on large datasets with SQL and built pipelines supporting client-specific model development and deployment on Google Cloud.

**Lead Mechanical Engineer – R&D** — Feb. 2019 – May 2021
*D&K Engineering* — *San Diego, CA*

- Led a 20-person team developing next-generation ecoATM recycling kiosks, earning a $1M contract extension; promoted to project lead.
- Built a suite of Python productivity tools (BOM cleaning, data logging, purchasing tracking) that raised in-house efficiency by roughly 50%.
- Developed a test fixture for Apple processors and delivered SolidWorks CAD/design support across hardware and medical programs, including Illumina (75% faster mold-swap tooling), Abbott, and a Stanford DNA-sequencer microfluidic system.

**Mechanical Engineer I** — Jan. 2017 – Jun. 2018
*Cobham Advanced Electronic Solutions* — *San Diego, CA*

- Designed and fabricated fixtures to GD&T standards for assembly and military-standard environmental testing of radar sensors used in defense systems.

**Flotilla Staff Officer / Member** — May 2015 – Present
*United States Coast Guard Auxiliary* — *San Diego, CA*

- Built Python productivity tools for the U.S. Coast Guard, including a pilot-training tracker adopted fleet-wide across all Air Stations nationwide; awarded the Coast Guard Auxiliary Achievement Medal.

## Projects

**Nahtadi – Islamic Prayer Times iOS App** | *Swift, SwiftUI, SwiftData, App Store* Aug. 2025 – Present

- Designed, built, and shipped a privacy-first iOS app (live on the App Store) computing the five daily prayer times and Qibla direction from astronomical algorithms, with zero data collection and full on-device operation.
- Architected in SwiftUI/SwiftData using MVVM, with multiple calculation methods, Hijri calendar, local notifications, and offline support.
- Implemented accessibility to WCAG standards: VoiceOver with phonetic Arabic labels, Dynamic Type, and reduced-motion support.

**Parallelizing Brent's Method with CUDA – M.S. Project** | *CUDA, C++, Python, GPU Computing, Nsight* Spring 2026

- Built the first known CUDA implementation of Brent's root-finding method, parallelizing 4M+ independent problems across GPU threads on an NVIDIA RTX 3080; benchmarked 35x kernel-level and 8.8x end-to-end speedup over a single-threaded C++ baseline.
- Engineered a Structure-of-Arrays memory layout for coalesced warp reads and enforced bit-identical results across Python, C++, and CUDA on 16,384 test cases via floating-point contraction controls.
- Profiled with NVIDIA Nsight Systems to isolate PCIe transfer and allocation as the dominant cost; specified pinned-memory and persistent-buffer optimizations projected to roughly double end-to-end throughput.

**Islamic Prayer Time Algorithm Library** | *Python, Scientific Computing, pytest, CI/CD* Jun. 2025 – Dec. 2025

- Built a zero-dependency, pure-Python scientific-computing library — the computational engine behind Nahtadi — implementing Julian Day calculations, spherical trigonometry, and high-latitude adjustments across 8+ calculation methods.
- Validated with 105 unit tests at 90%+ coverage and continuous integration via GitHub Actions.

**Mini Compiler** | *Python, LL(1) Parser, AST, pytest, GitHub Actions, CI/CD* Aug. 2024 – Dec. 2024

- Expanded a 339-line academic project into a 4,000+ line modular compiler translating a Pascal-like language to executable Python, with a table-driven LL(1) predictive parser, semantic analysis, and AST-based code generation.
- Includes 78 tests at 71% coverage, complete type hints, zero external dependencies, and GitHub Actions CI/CD.

**Maritime Collision Avoidance Training System** | *Python, Streamlit, NumPy, pytest* May 2024 – Present

- Built an educational web app — in active use by the U.S. Coast Guard Auxiliary — that computes Closest Point of Approach and course/speed maneuver solutions from radar observations using vector-based relative-motion algorithms.
- Implemented in Python/Streamlit with full pytest coverage and type-checked, linted code (MyPy, Ruff).

## Technical Skills

**Languages:** Swift, Python, C++/C, SQL
**iOS/Apple:** SwiftUI, SwiftData, Xcode, XCTest, MVVM, Accessibility (VoiceOver, Dynamic Type), App Store Publishing
**ML/Scientific Computing:** PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Computer Vision, CUDA, Numerical Methods
**Tools/DevOps:** Git, GitHub Actions, CI/CD, pytest, MyPy, Ruff, Docker, Nsight
**Cloud:** AWS (Lambda, DynamoDB, S3), Google Cloud
**Databases:** MySQL

## Certifications

**Cloud:** AWS Knowledge: Serverless (AWS); Google Cloud Computing Foundations (Google Cloud)
**Academic:** Algorithms: Design and Analysis, Part 1 (Stanford Online); Data Science Immersive — 500+ hrs (General Assembly)
**Engineering:** Certified SolidWorks Associate – CSWA (Dassault Systèmes)